

Closed-Form Evaluation of Correlated PFS and OS Group Sequential Designs

From joint correlation theory to a reusable design evaluator

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SECTION 01

Project overview

The design setting, the unresolved calculation, and our deliverable

Phase 3 oncology trials combine two related time-to-event endpoints

Two endpoints

PFS is the earlier disease-control endpoint.

OS is the later survival endpoint.

Both are observed in the same randomized trial.

Sequential design

Group-sequential interim analyses are triggered by events.

This presentation uses the default PFS→OS sequence: OS is tested only after PFS success.

This project closes a reusable correlated-design gap

Missing calculation

Existing tools support single-endpoint boundaries, but not the joint PFS–OS success probability across looks.

Closed-form evaluator

Derive moments and correlation matrix in closed form, then use multivariate normal integration to compute marginal and joint power.

Reusable delivery

Use clinically interpretable inputs: medians, hazard ratios, accrual, and event targets and expose the workflow through an R package and Shiny application.

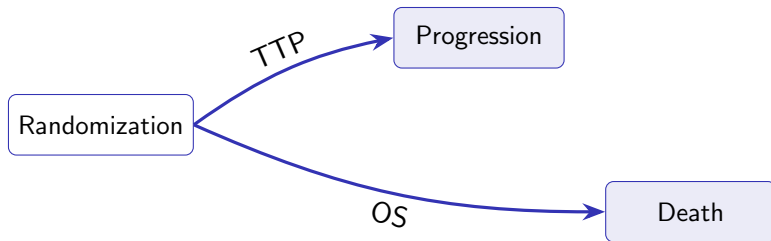
Theory → Closed-form evaluator → R package and R Shiny app

SECTION 02

Trial design context

How the PFS-to-OS decision is defined

PFS is the first event among TTP and OS

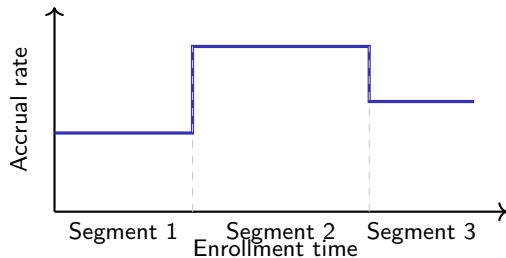


Fleischer relationship

$$PFS = \min(TTP, OS)$$

The first of progression and death defines PFS, while OS ends at death; this shared event history induces dependence.

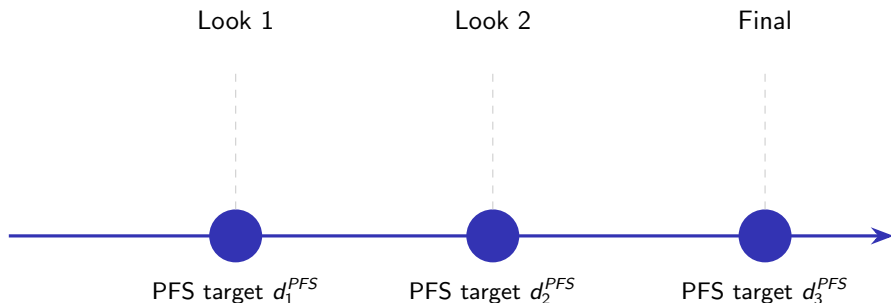
Piecewise-uniform accrual connects enrollment to event-driven analysis



Design implication

Conditional on planned segment counts, enrollment time has a piecewise-uniform density.

Event-driven group-sequential design creates multiple look times



At each look time A_ℓ , PFS and OS are evaluated on the same calendar cutoff time. Closed-form integrals are summed across segments, while each analysis is triggered by a target number of PFS events.

Each look produces statistics, but their joint distribution is unknown

What the design produces

For endpoint $k \in \{PFS, OS\}$ and look $\ell \in \{1, \dots, L\}$, the event-driven analysis produces test statistics Z_ℓ^{PFS} and Z_ℓ^{OS} . Therefore we have a $2L$ -dimensional vector of statistics $\mathbf{Z} = (Z_1^{PFS}, \dots, Z_L^{PFS}, Z_1^{OS}, \dots, Z_L^{OS})^\top$.

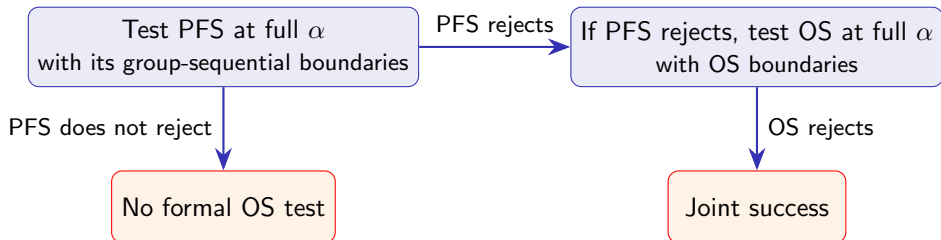
The missing design input

The statistics can be generated, but the joint distribution is not yet known. The subject-level event and follow-up vectors satisfy a multivariate CLT; the Delta method then gives an asymptotic normal law for $\mathbf{Z}$. We need to derive:

- Dependence within an endpoint.
- Dependence between PFS and OS.

The central problem is to derive the joint distribution of $\mathbf{Z}$.

PFS-to-OS gatekeeping defines the formal success path



This presentation illustrates the default PFS→OS sequence. The package and Shiny app also support the reverse fixed OS→PFS sequence.

Scope and limitations

Included

- Piecewise-uniform accrual.
- Event-driven, synchronized analyses.
- Fixed testing sequence.

Limitations

- No dropout.
- No graphical alpha recycling.
- No co-primary success rule.

SECTION 03

Closed-form correlation engine

Covariance and variance first; calendar cutoff last

Basic notation: Fleischer event times and estimators

Fleischer model

$$TTP \sim \text{Exp}(\lambda_1),$$

$$OS \sim \text{Exp}(\lambda_2).$$

$$TTP \perp OS, \quad PFS = \min(TTP, OS),$$

$$F_{i,\ell} = \max(0, A_\ell - U_i),$$

$$\tilde{T}_{i,\ell}^k = \min(T_i^k, F_{i,\ell}),$$

$$\delta_{i,\ell}^k = \mathbf{1}(T_i^k \leq F_{i,\ell}).$$

Follow-up, totals, and estimators

$$\hat{\theta}_\ell^k = \log \left(\frac{D_{T,\ell}^k}{S_{T,\ell}^k} \right) - \log \left(\frac{D_{C,\ell}^k}{S_{C,\ell}^k} \right).$$

$$Z_\ell^k = -\frac{\hat{\theta}_\ell^k - \theta_0^k}{\text{se}(\hat{\theta}_\ell^k)}.$$

where $D_{g,\ell}^k$ and $S_{g,\ell}^k$ are cumulative truncated events and follow-up for group $g \in \{T, C\}$, endpoint $k \in \{PFS, OS\}$ and $\ell \in \{1, \dots, L\}$.

CLT and Delta method yield the joint distribution of $\mathbf{Z}$

Per-subject vector from the technical report

CLT input. For group g and looks ℓ_1, ℓ_2 , define

$$X_{i,g} = \left(\delta_{i,\ell_1}^{PFS}, \tilde{T}_{i,\ell_1}^{PFS}, \delta_{i,\ell_2}^{OS}, \tilde{T}_{i,\ell_2}^{OS} \right)^\top.$$

$$\mu_X^{(g)} = \mathbb{E}[X_{i,g}], \quad \Sigma_X^{(g)} = \text{Cov}(X_{i,g}).$$

$$\sqrt{n_g} \left(\bar{X}_g - \mu_X^{(g)} \right) \xrightarrow[n_g \rightarrow \infty]{d} \mathcal{N}_4 \left(0, \Sigma_X^{(g)} \right).$$

Delta method: map to the log-rate statistics

Delta-method map. With

$$h(x) = \left(\log \frac{x_1}{x_2}, \log \frac{x_3}{x_4} \right)^\top,$$

Delta method: $\Sigma_h^{(g)} = J \Sigma_X^{(g)} J^\top$.

$$\sqrt{n_g} \left(h(\bar{X}_g) - h(\mu_X^{(g)}) \right) \xrightarrow{d} \mathcal{N}_2 \left(0, \Sigma_h^{(g)} \right).$$

After combining arms and standardizing:

$$\mathbf{Z} \xrightarrow{d} \mathcal{N}_{2L}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

The Delta method reduces covariance and variance to moments

Delta-method output

$$\text{Cov}(\hat{\theta}_a, \hat{\theta}_b) \approx J_a \Sigma_X J_b^\top,$$

$$\text{Var}(\hat{\theta}_a) \approx J_a \Sigma_X J_a^\top.$$

J_a is the gradient of the log-rate map. Once Σ_X is known, every covariance and variance term follows mechanically.

Only moments are numerical targets

$$\mu_X = \mathbb{E}[X], \quad \Sigma_X = \text{Var}(X).$$

The closed-form engine therefore needs only expectations and variances (equivalently, second moments) of the Appendix X vector.

Cross moments are closed form after accrual averaging

Step 1: representative conditional product

For $a_1 = A_{\ell_1} - u \leq a_2 = A_{\ell_2} - u$,

$$\begin{aligned}\mathbb{E}\left[\delta_{\ell_1}^{PFS} \delta_{\ell_2}^{OS} \mid a_1, a_2\right] \\ = 1 - e^{-\lambda_2 a_2} - e^{-\lambda_1 a_1} + e^{-\lambda_1 a_1 - \lambda_2 a_2}.\end{aligned}$$

The other three cross products use the same shared (*TTP*, *OS*) event-time structure.

Step 2: average over entry time

$$\begin{aligned}m_X^{(g)}(A) &= \frac{1}{R} \int_0^{\min(A, R)} \\ &\quad \mathbb{E}[X \mid F = A - u], du.\end{aligned}$$

The integrands are finite combinations of exponential-polynomial terms, so every required moment is closed form.

Moment output

The covariance and variance formulas now reduce to closed-form per-subject moments.

Calendar cutoffs close the event-driven construction

Expected PFS target determines A_ℓ^*

$$n_T m_{\delta PFS}^{(T)}(A_\ell^*) + n_C m_{\delta PFS}^{(C)}(A_\ell^*) = d_\ell^{PFS}.$$

The left-hand side is strictly increasing, so each look has one deterministic numerical root. We use this expected-value cutoff in all subsequent moment and correlation calculations.

Synchronized analysis

OS uses the same synchronized calendar cutoff A_ℓ^* at each look. Its expected event count is evaluated at that cutoff.

Anscombe's theorem: a large-sample cutoff approximation

Anscombe's theorem: replacing a random stopping time

As the total sample size n grows, suppose the random operational time converges in probability to the deterministic expected cutoff,

$$\hat{A}_\ell \xrightarrow{P} A_\ell^*.$$

Under Anscombe's regularity conditions, the statistic evaluated at $\hat{A}_\ell$ has the same large-sample distribution as at A_ℓ^* . The estimator under expected cutoffs converge in distribution to the same multivariate normal limit as the estimator under random cutoffs as the sample size grows to infinity:

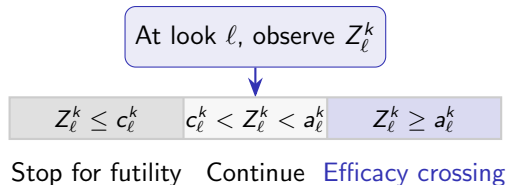
$$\mathbf{Z}_n(A^*) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}_{2L}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

SECTION 04

From joint distribution to decisions

Boundaries, gatekeeping, and joint power

Non-binding futility can stop an endpoint early



Default futility boundary

$$c_\ell^k = -\log\left(\text{HR}_{\text{futility}}^k\right) \sqrt{d_\ell^k \frac{r}{(1+r)^2}}.$$

The default $\text{HR}_{\text{futility}} = 1.2$ represents an unfavorable effect. With no futility rule, $c_\ell^k = -\infty$.

Non-binding means

Early stopping for futility does not affect the calculation of the efficacy boundaries.

gsDesign computes the three efficacy boundary families

Information used by gsDesign

$$t_{\ell}^{PFS} = \frac{d_{\ell}^{PFS}}{d_L^{PFS}}, \quad t_{\ell}^{OS} = \frac{d_{\ell}^{OS}}{d_L^{OS}}.$$

PFS information uses expected event counts at each look. Synchronized OS event counts are obtained from the closed-form moment calculations.

Three spending functions

```
sfu = gsDesign::sfLDOF  
sfu = gsDesign::sfLDPocock  
sfu = gsDesign::sfHSD
```

OF, Pocock, and HSD correspond to the three supported efficacy-boundary families.

Marginal and joint power

Marginal power

First crossing at look 1

First crossing at look 2

:

First crossing at look L

Each blue box evaluates:

$$\Pr\left(c_h^k \leq Z_h^k \leq a_h^k, h < \ell; Z_\ell^k \geq a_\ell^k\right).$$

Total marginal power is the sum of the probabilities in this column.

Joint power

		OS first crossing, j			
		1	2	...	L
PFS first crossing, i	1	E	E	...	E
	2	-	E	...	E
	...	-	-	...	...
	L	-	-	-	E

For $j \geq i$,

$$E_{ij} = \left\{ \begin{array}{ll} c_h^{PFS} \leq Z_h^{PFS} \leq a_h^{PFS} \ (h < i), & Z_i^{PFS} \geq a_i^{PFS}, \\ c_h^{OS} \leq Z_h^{OS} \leq a_h^{OS} \ (i \leq h < j), & Z_j^{OS} \geq a_j^{OS} \end{array} \right\}.$$

Total joint power is the sum of $\Pr(E_{ij})$ over all valid upper-triangular cells $j \geq i$.

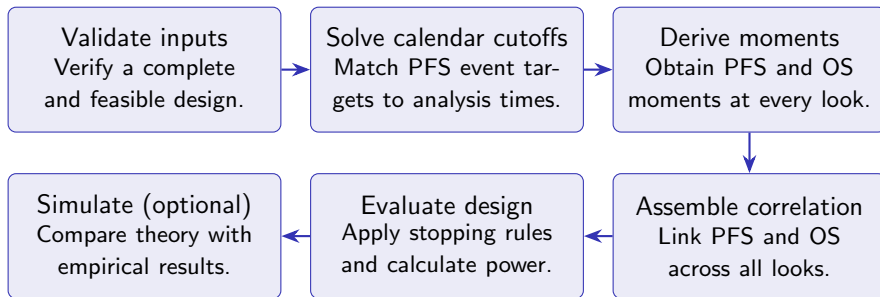
SECTION 05

Implementation and validation

From reusable software to empirical operating characteristics

One pipeline from design settings to results

Each stage converts clinical design assumptions into decision-ready quantities.



The Shiny interface exposes the same pipeline through design inputs, boundary summaries, power results, and optional simulation validation.

Live demo: begin with a simple prefilled two-look design

Total sample size	200
Allocation	1 : 1
PFS event looks	50, 100
Accrual rate	20 participants/month
Accrual duration	10 months
Control PFS / OS median	4/8 months
PFS / OS HR	0.80/0.85
Testing order	PFS→OS
Efficacy spending	OF / OF

Live changes

- Start with final-only efficacy and no futility.
- Enable efficacy at both looks.
- Add first-look futility.
- Compare boundaries and power output.

Accuracy study: scale the same three-look design across three sample sizes

Total n	Treatment / control	PFS look 1	PFS look 2	PFS look 3
400	200 / 200	120	180	240
1,200	600 / 600	360	540	720
2,000	1,000 / 1,000	600	900	1,200

Study question

Across three sample sizes, how closely do the theoretical operating characteristics agree with empirical estimates of correlation, incremental marginal power, and each PFS-to-OS joint success path?

The accuracy comparison holds the underlying design assumptions fixed

Event model

Control medians: 6-month PFS and 18-month OS.

Treatment HRs: 0.75 for PFS and 0.80 for OS.

Sequential design

PFS: Pocock.

OS: O'Brien–Fleming.

PFS futility at looks 1 and 2 only.

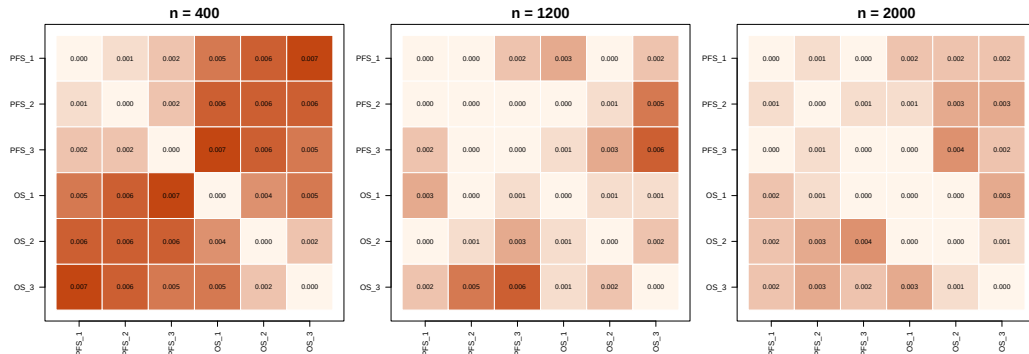
Reproducibility

50,000 replicates per size.

Integration seed: 777.

Simulation seed: 20260818.

Theoretical and empirical correlations agree closely



Readout

Each cell is the absolute theoretical-empirical correlation difference; each panel uses 50,000 empirical simulation replicates.

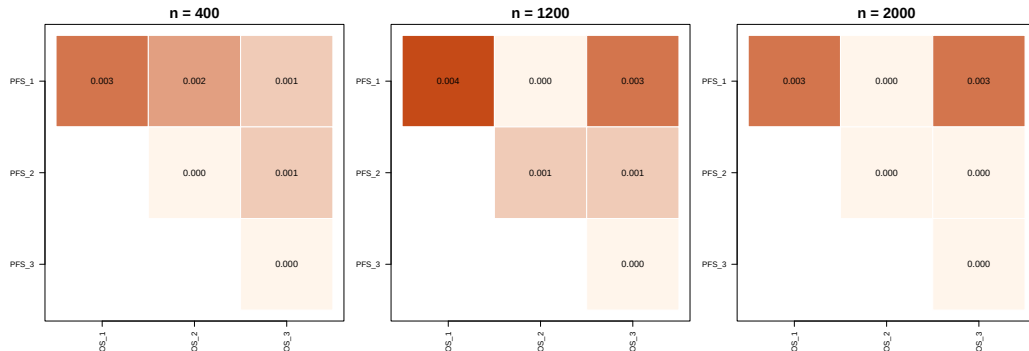
Theoretical and empirical incremental marginal power agree look by look



Readout

Each cell is the absolute theoretical-empirical incremental marginal power difference; each panel uses 50,000 empirical simulation replicates.

Theoretical and empirical joint power agree path by path



Readout

Each cell is the absolute theoretical-empirical joint power difference; each panel uses 50,000 empirical simulation replicates.

Theoretical and empirical power complete the accuracy readout

Total n	PFS cumulative power			OS cumulative power			Total joint power		
	Theoretical	Empirical	$ \Delta $	Theoretical	Empirical	$ \Delta $	Theoretical	Empirical	$ \Delta $
400	0.5289	0.5248	0.0041	0.2218	0.2158	0.0060	0.1738	0.1682	0.0056
1200	0.9502	0.9491	0.0012	0.5444	0.5428	0.0015	0.5366	0.5348	0.0017
2000	0.9970	0.9968	0.0002	0.7633	0.7623	0.0009	0.7628	0.7619	0.0009

Accuracy conclusion

Our closed-form engine produces theoretical results that closely match empirical simulation across correlation, marginal power, and joint power. The agreement improves with increasing sample size, as expected from the large-sample CLT.

SECTION 06

Summary and outlook

What is delivered, what is established, and what comes next

The GitHub package is the reproducibility and reference entry point

Available in the repository

- R package source, tests, and Shiny application.
- Methodology vignette and worked examples.
- Reproducible analysis scripts and documented assumptions.
- Full literature references supporting the model and testing framework.

<https://github.com/xingjian-ma/CorrSurvGSD>

Takeaway: correlated group-sequential evaluation is now reproducible

- 1 Closed-form per-subject moments supply the missing cross-endpoint correlation block.
- 2 The resulting joint normal distribution supports boundary-aware marginal and joint power calculations.
- 3 Empirical evidence supports the theoretical operating-characteristic calculations across increasing design sizes.
- 4 The package and Shiny interface make the workflow accessible for future design exploration.

A transparent route from design assumptions to correlated operating characteristics.

The current engine creates a clear path to three extensions

Alpha allocation

$$\alpha/2 \quad \alpha/2$$

Separate endpoint boundaries, then enumerate the permitted testing paths.

Random dropout

Censoring process

Extend the moment calculation to non-informative dropout.

Co-primary endpoints

$$\text{PFS} \cap \text{OS}$$

Replace sequential gatekeeping with a joint-success criterion.

References

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- 2 Hung, H. M. J., Wang, S.-J., and O'Neill, R. (2007). "Statistical considerations for testing multiple endpoints in group sequential or adaptive clinical trials." *Journal of Biopharmaceutical Statistics*, 17, 1201–1210.
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Questions & Discussion